# Free-fall lab

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Estimate g from measured motion and discuss measurement limits.

**SUCCESS CRITERIA - I can...**

1. Keep supplied examples separate from measurements.
2. Find acceleration from a velocity-time slope.
3. Use repeated trials and explain uncertainty.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “slope” and “trial” to explain your model or result. Sentence starters are optional.

**slope**: vertical change divided by horizontal change on a graph

**trial**: one run or repetition of a test

**acceleration**: rate of change of velocity

## 01 Bring back an earlier idea

A supplied gravity model has $Gm_1m_2=75$ N·m² and separation 5 m.
**Find** the force and its direction.

**Word help:** separation = distance between the specified objects or centers

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $a = \Delta v/\Delta t$; $v_f = v_i + a\Delta t$.
Velocity-time slope estimates acceleration. Downward is positive. Use the approved apparatus; distinguish supplied from measured data.

**Worked reasoning:** Supplied trial values 8, 10, 12 m/s² have mean 10 m/s² and range 8-12. The spread describes variation; averaging does not remove every systematic error.

**Your turn:** Plot the three values on a number line and mark the mean.
**Name** an error that could shift all trials together.

**Sentence starter:** The step I would change is ... . The evidence is ... .

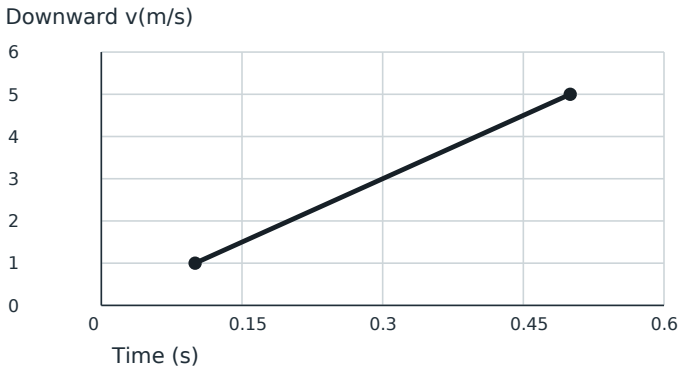
**Marzano check - circle:** ① ② ③

## 03 Look and explain

**Use** the supplied velocity-time graph to estimate acceleration.
**Show** the change in velocity and elapsed time.

**Word help:** estimate = find an approximate value

**Sentence starter:** I found ... [unit]. This value means ... in this situation.




**Marzano check - circle:** ① ② ③

## 04 Build the solution

Three supplied trial estimates are 9, 10 and 11 m/s².
**Find** their mean and range. Does variation mean the experiment is useless?

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Use the idea

Plan three repeated trials.
**Name** a controlled condition, a measurement you will record and a reason not to include a badly pushed release without comment.

**Sentence starter:** My design shows … . It meets the requirement by … .

**Marzano check - circle:** 1 2 3

## 06 Choose a method

A box at rest has 18 N right and 4 N left.
**Find** the additional horizontal force needed for balance and show the sum.

**Word help:** at rest = not moving in the chosen reference frame

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** 1 2 3

## 07 Exit check

An interval changes downward velocity from 4 to 6 m/s in .4 s.
**Find** a and name one source of uncertainty.

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: Apply it another way

Two groups have values 9, 10, 11 and 5, 10, 15 m/s².
**Compare** means and repeatability without declaring either perfectly accurate.

**Sentence starter:** Both … . They differ in … because … .

**Marzano check - circle:** 1 2 3